SiT Dataset : Socially Interactive Pedestrian Trajectory Dataset for Social Navigation Robots

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Github Homepage

Contributions

✓ **Static collection**
- Many pedestrian trajectory datasets were collected from fixed positions, limiting variability.

✓ **Missing ego motion**
- Many existing robot datasets lack ego-motion information.

✓ **Vehicle-centric bias**
- Most autonomous datasets were collected from vehicles, exhibiting a gap from realistic robot-pedestrian interaction behavior.

*†: Multi – layered map

Dataset	Platform	Task	Sync.	Map	E2E	Location
UCY	Fixed	Tracking, Prediction	-			Outdoor
ETH	Fixed	Tracking, Prediction	-			Outdoor
SDD	Fixed	Tracking, Prediction	-			Outdoor
CITR-DUT	Fixed	Tracking, Prediction	-			Outdoor
nuScenes	Vehicle	Detection, Tracking, Prediction	-	✓†		Outdoor
Waymo Open	Vehicle	Detection, Tracking, Prediction	✓	✓		Outdoor
Argoverse	Vehicle	Detection, Tracking, Prediction	✓	✓†	✓	Outdoor
JRDB	Robot	Detection, Tracking				Indoor & Outdoor
STCrowd	Fixed	Detection, Tracking, Prediction	✓			Outdoor
SiT(Ours)	**Robot**	**Detection, Tracking, Prediction**	✓	✓†	✓	**Indoor & Outdoor**

Overview

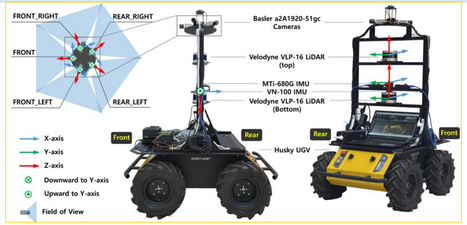
✓ **Real-world context**
- Collected data from dense areas like campuses and public roads

✓ **Comprehensive data**
- Sequential raw data from various sensors
- 60 scenes with 60K images and 12K point cloud frames at 10 Hz
- 2D and 3D bounding boxes for 6 classes

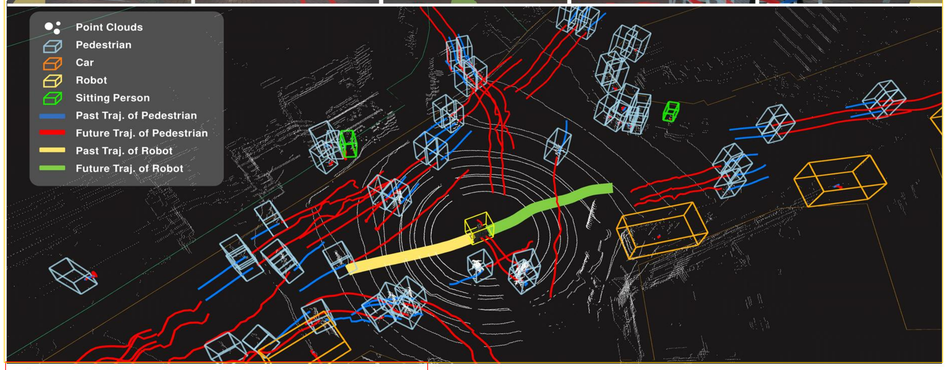
✓ **Unique features**
- Emphasis on *Human-Robot Interactions* (HRI)
- Precise multi-sensor synchronization
- Multi-layered indoor & outdoor semantic maps from SLAM
- Cover tasks from 3D detection to motion forecasting (end-to-end)

Robot Setup

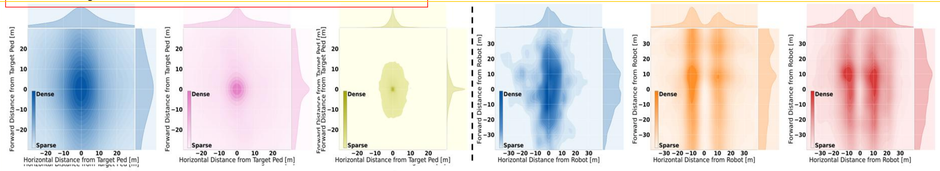
✓ **Multi-sensors equipped on UGV**
- 1 x Husky UGV platform
- 2 x Velodyne VLP-16
- 5 x Basler a2A1920-51gc cameras
- 1 x MTi-680G IMU & RTK
- 1 x VectorNav VN-100 IMU



Details of SiT Dataset

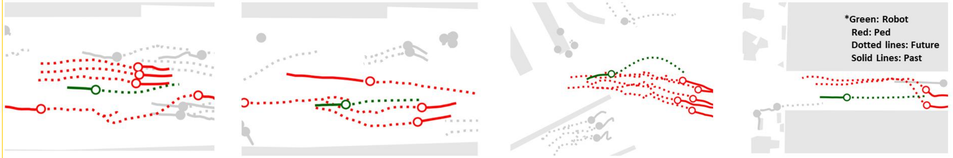


✓ **Comparison with other datasets**



SiT(Ours) SSD ETH/UCY SiT(Ours) Waymo Open nuScenes

✓ **Examples of human-robot interactions**



Approach Followed by Pedestrian Avoidance by Robot Avoidance by Pedestrians

Tasks and Benchmark

✓ **Perception tasks**
- 3D Object Detection based on image or point clouds
- 3D Multi-Object Tracking
- Trajectory Prediction
- End-to-End 3D Detection to Motion Forecasting

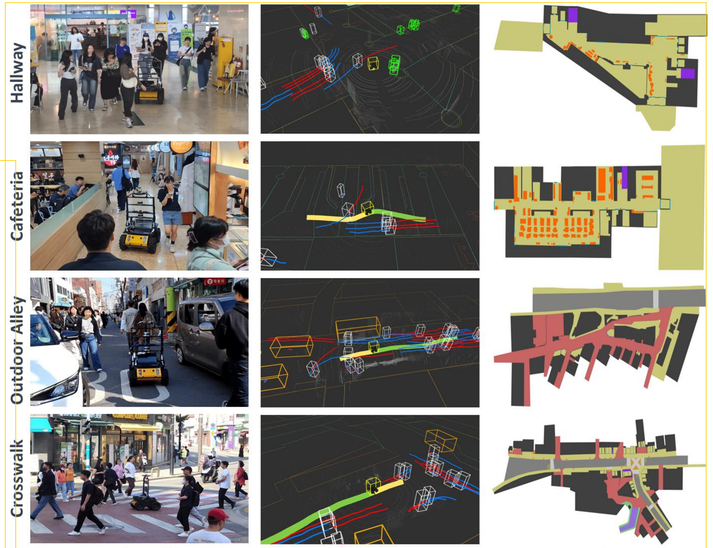
✓ **Benchmark**
- Plan to release benchmarks and pre-trained models.
- Challenges open on **Eval.AI** *(Feb. 2024)*

Samples of dataset

✓ **Real-world captures**
- Candid shots from diverse urban environments

✓ **3D visualization**
- Dynamic representations of objects and trajectories

✓ **Semantic maps**
- 12 Color-coded layouts detailing regions



Real World 3D Visualization Semantic Map